# Chapter 5

# The Technological Framework and Software Stack

## 5.1 Backend Infrastructure and Data Platform Architecture

To operationalize the domain-oriented Data Product model and support interactive analytical workloads within Amadori S.p.A., the enterprise data architecture is undergoing an evolutionary transition from an on-premises relational infrastructure toward a scalable, cloud-native data platform.

The baseline enterprise environment relies primarily on an on-premises Oracle database, which historically serves as the operational transaction repository and the target for analytical queries. To overcome the computational constraints of centralized on-premises storage and eliminate contention between transactional processing and resource-intensive analytical calculations, the target architecture introduces Google BigQuery as the enterprise cloud data warehouse. BigQuery provides a serverless, highly scalable environment based on distributed compute engines and columnar storage, enabling parallelized execution of complex analytical queries over large-scale historical datasets.

Rather than implementing an abrupt cutover, the transition follows a phased migration strategy. In the initial phase, analytical assets and ingestion pipelines continue to read directly from the operational Oracle database. As data models, historical partitions, and ETL routines are progressively refactored and validated, tables are transferred to Google BigQuery. During this transitional period, both database connections remain active simultaneously.

The presence of a centralized Semantic Layer implemented in Looker plays a strategic role in facilitating this dual-connection phase. Because Looker functions via in-database querying rather than data extraction, it can maintain concurrent

dialect connections to both Oracle and BigQuery. Business dimensions and metrics formalized in LookML can initially target tables hosted on the Oracle instance; subsequently, as individual domains complete their migration to BigQuery, the underlying connection references within the LookML models are updated to point to the cloud tables. This decoupling ensures that downstream analytical consumers, executive reports, and AI interfaces remain completely shielded from the physical migration process, experiencing no disruption in metric definitions or query behavior while backend storage is shifted.

## 5.2 Formalizing Business Data: The Semantic Layer with Looker and LookML

To operationalize the conceptual consensus established by the corporate ontology and business glossary, the technological framework leverages Looker and its proprietary modeling language, LookML (Looker Modeling Language). Within the updated data stack, Looker does not function merely as a visualization tool; rather, it serves as the definitive, centralized execution engine for the enterprise Semantic Layer. By shifting the responsibility of SQL generation away from individual business analysts and fragmented reporting tools, LookML acts as a version-controlled, object-oriented code layer that abstracts physical database complexity into standardized business dimensions and measures. This implementation precisely mirrors the architectural recommendations of Hilger, Tesfaye, and Wahl [10], establishing an un-bypassable governance gateway that translates raw data lakehouse tables into certified, consumption-ready enterprise intelligence.

The LookML architecture is structured around two core abstractions: *Views* and *Explores*. Inside LookML views, physical database columns are mapped directly to logical fields. It is within this declarative code environment that the strict formulas governing critical supply chain metrics are codified. For example, the complex business logic required to isolate the *Consuntivo Ricondizionato* (the smoothed historical sales baseline, free from promotional spikes) and calculate the 33% promotional threshold is written exactly once within the view layer. Similarly, the multi-level hierarchical Bill of Materials (BOM) logic—which governs the relationship between the abstracted *Neutral Article* and finished, customer-facing stock-keeping units (SKUs)—is declared programmatically. LookML *Explores* then define the semantic join relationships between these views, mirroring the multi-layered supply chain grid. An explore dynamically models how a *Forecast Client* node connects to downstream *Shipping Clients*, enforcing the explicit lookback windows and filtering metrics required for precise daily demand allocation.

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The LookML architecture maps dimensions, measures, aggregates, and relationships through a developer-friendly, text-based syntax [18]. This code-first methodology represents an agile development pattern that stands in sharp contrast to traditional drag-and-drop, metadata-driven business intelligence solutions that require high upfront modeling investments and suffer from manual maintenance bottlenecks. By defining enterprise rules—such as the mathematical calculations driving the smoothed historical baseline and the explicit 20-week lookback perimeters for shipping allocations—directly inside LookML views, technical data teams can generate highly optimized SQL queries at runtime on demand [18]. Furthermore, because LookML models are written as plain text code, they integrate natively into standard software engineering lifecycles [18]. Definitions are managed via centralized Git repositories, enabling code versioning, branch testing, and automated continuous integration (CI/CD) pipelines before metrics are compiled and served as a unified API to downstream consumers [18].

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This code-driven approach to data semantics delivers substantial operational benefits. Because the LookML repository is managed via standard Git workflows, any alteration to a corporate metric definition must undergo strict peer review, automated testing, and CI/CD validation before being deployed into production. This architecture effectively eliminates the risk of metric drift, ensuring that whether a query is generated by a human supply chain planner, an automated materials procurement pipeline, or an external optimization algorithm, the underlying SQL execution remains perfectly identical. Ultimately, Looker transforms

the abstract corporate ontology into a live, discoverable, and programmatically queryable API layer, providing the deterministic semantic runtime required to support advanced cognitive systems.

## 5.3 Governance, Glossary, and the Corporate Knowledge Graph: The OpenMetadata Ecosystem

While Looker provides the execution engine for data semantics, maintaining comprehensive visibility, auditable data contracts, and automated lineage across a decentralized Data Product architecture requires an enterprise-grade metadata platform. This capability is realized through the deployment of the OpenMetadata ecosystem. OpenMetadata is an open-source, schema-first metadata management platform that utilizes standardized JSON schemas to unify data discovery, governance, and quality monitoring into a single, cohesive repository. By treating metadata as a dynamic, operational asset rather than static documentation, OpenMetadata provides the structural transparency required to enforce the data governance frameworks outlined in the DAMA Data Management Body of Knowledge [5].

Within this architecture, OpenMetadata serves as the unified orchestrator for the enterprise *Business Glossary*. Rather than maintaining definitions in isolated text documents, the glossary terms are directly injected into OpenMetadata's core catalog and programmatically linked to live technical assets. A glossary term such as "Carcass Mass-Balance" or "Neutral Product Allocation" is explicitly mapped to the exact LookML views, database schemas, and data pipelines that compute those entities. This deep integration establishes total end-to-end data lineage, allowing data stewards to trace the complete evolutionary path of an informational asset. If an upstream transactional schema within the primary ERP changes or an anomaly occurs during the automated weekly rolling forecast execution, OpenMetadata's automated ingestion daemons instantly detect the mutation, perform an impact analysis across the dependency web, and notify downstream data consumers before the structural modification can corrupt corporate planning models.

Furthermore, OpenMetadata abstracts these interconnected technical components and metadata relationships to construct a live, dynamic corporate *Knowledge Graph*. Every entity in the data ecosystem—ranging from raw storage buckets and data lakehouse tables to Looker dimensions, machine learning pipelines, and human organizational roles—is represented as a distinct node within a highly structured graph database topology. These nodes are bound together by directed semantic edges that explicitly define their relationships, dependencies, and operational ownership. For example, a specific finished SKU node is linked via an

ontological edge to its parent Neutral Article node, which is connected to its certifying Data Champion and bound by a specific data contract SLA. By unifying master data, technical metadata, and business governance definitions into an operational graph, the enterprise can programmatic evaluate data quality by design, ensuring absolute data trustworthiness and providing a context-rich roadmap to guide autonomous artificial intelligence engines.

## 5.4 Component Integration and Orchestration for Generative AI Enablement

The final, unifying layer of the technological framework is the orchestration and integration architecture that connects the cloud data platform, the Looker Semantic Layer, and the OpenMetadata Knowledge Graph to enable an intuitive, conversational Generative AI analytical experience. Historically, deploying Large Language Models (LLMs) over enterprise databases required models to write raw SQL code directly against physical schemas—a design choice that frequently leads to catastrophic query errors, unhandled table joins, and severe logical hallucinations. The integrated stack presented here completely bypasses this vulnerability by introducing a structured Graph-Augmented Retrieval (GraphRAG) loop, ensuring that autonomous AI agents interact exclusively with certified semantic models rather than raw, ungoverned data infrastructures.

The orchestration sequence begins when a corporate supply chain planner inputs a natural language query into the user interface (e.g., "What is our projected organic baseline demand for fresh whole birds over the next four weeks, excluding scheduled retail promotions?"). This prompt is intercepted by a specialized AI Agent framework, which serves as the cognitive orchestrator of the ecosystem. Instead of directly executing next-token probability distributions over the user's raw text, the AI Agent transmits the query to the OpenMetadata API to resolve structural and semantic context. OpenMetadata interrogates the corporate Knowledge Graph and returns a tailored context packet containing the certified definitions from the Business Glossary, the active data contract SLAs, and the precise object paths of the relevant semantic entities.

Armed with this validated ontological context, the AI Agent interacts directly with the Looker Semantic Layer via the Looker SDK, translating the user's natural language intent into a structured Looker API call rather than an arbitrary SQL string. The Semantic Layer processes the request, safely executes the pre-compiled, optimized LookML queries against the backend lakehouse storage engine, and returns mathematically validated, deterministic metrics back to the agent. Finally, the AI Agent passes these verified numbers, alongside the lineage metadata,

through the LLM to generate a natural, context-aware, and completely verifiable response for the end user.

Figure 5.1: The Cognitive GraphRAG Execution and Verification Loop

Through this sophisticated integration and orchestration loop, the enterprise successfully resolves the deep-seated epistemic friction between data writing and data reading. By forcing the Generative AI engine to route all analytical requests through a centralized Semantic Layer and a governed Knowledge Graph, the architecture establishes an unyielding barrier against algorithmic hallucinations. Planners are empowered to conduct complex supply chain simulations, execute multi-level BOM explosions, and isolate baseline trends from promotional uplifts through fluid conversational interfaces, secure in the knowledge that every output is mathematically certified by the underlying data platform.

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